Evaluating the Economic Impact of Climate Policy through Emissions, Energy, and Taxation

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This report analyzes climate policy and environmental outcomes from 2016 to 2021—a period marked by international climate commitments such as the Paris Agreement, national carbon reduction targets, and a global push for clean energy. It also includes the disruptive impact of the COVID-19 pandemic in 2020. By examining environmental tax, renewable energy use, and emissions trends, the report evaluates how countries advanced sustainability during a time of both progress and crisis.

Data Sources:

- **OECD Greenhouse Gas Emissions Dataset:**
  https://dataexplorer.oecd.org/vispg=0&bp=true&snb=33&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_AIR_GHG%40DF_AIR_GHG&df[ag]=OECD.ENV.EPI&df[vs]=1.0&dq=.A.GHG..KG_CO2E_PS&pd=2014%2C&to[TIME_PERIOD]=false&isAvailabilityDisabled=false&hc[Measure]=&tm=Air%20emissions
- **OECD Renewable Energy Consumption Dataset:**
  https://dataexplorer.oecd.org/vispg=0&bp=true&snb=32&vw=tb&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_SDG%40DF_SDG&df[ag]=OECD.WISE.RSB&df[vs]=1.0&dq=..7_2%2B1_2..._T._T._T._T._T.&to[TIME_PERIOD]=false&lom=LASTNPERIODS&lo=5&isAvailabilityDisabled=false&hc[SDG%20goal]=&tm=Sustainable%20Development%20Goals
- **OECD Environmental Tax Revenue Dataset:**
  https://dataexplorer.oecd.org/vispg=0&bp=true&snb=23&vw=tb&df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_ERTR%40DF_ERTR&df[ag]=OECD.ENV.EPI&df[vs]=1.0&dq=A..TAXREV.CAT_ENE._T..&pd=2017%2C&to[TIME_PERIOD]=false&isAvailabilityDisabled=false&hc[Unit%20of%20measure]=&hc[Category%20of%20environmentally%20related%20tax%20revenue]=&tm=Environmentally%20related%20tax%20

Tracking the Greenhouse Effect: Emission Patterns Over Time

A cross-country analysis of GHG emissions from 2016 to 2021.

(CTRL + Click) while using buttons!

Filter

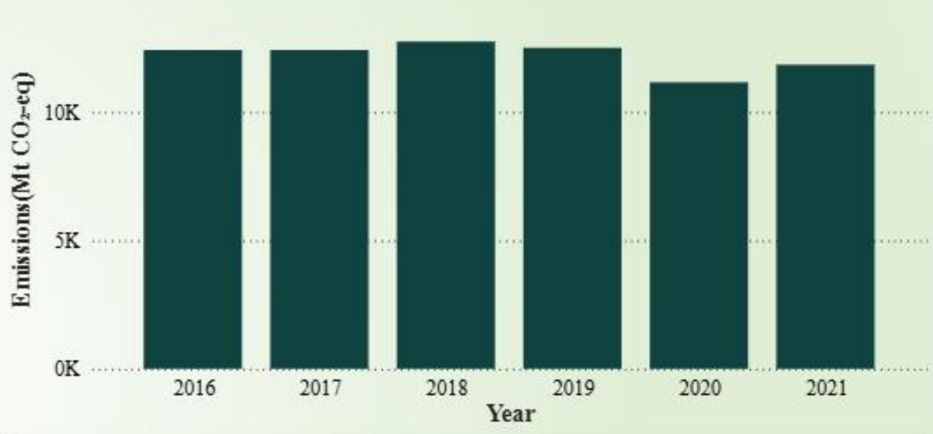
Country-wise Emissions

Country	Emissions (Mt CO_2-eq)
United States	66,300.28
United Kingdom	4,858.92
Belgium	1,184.88
Portugal	652.35
Lithuania	154.71
Total Emissions (Mt CO_2-eq)	**73,151.14**

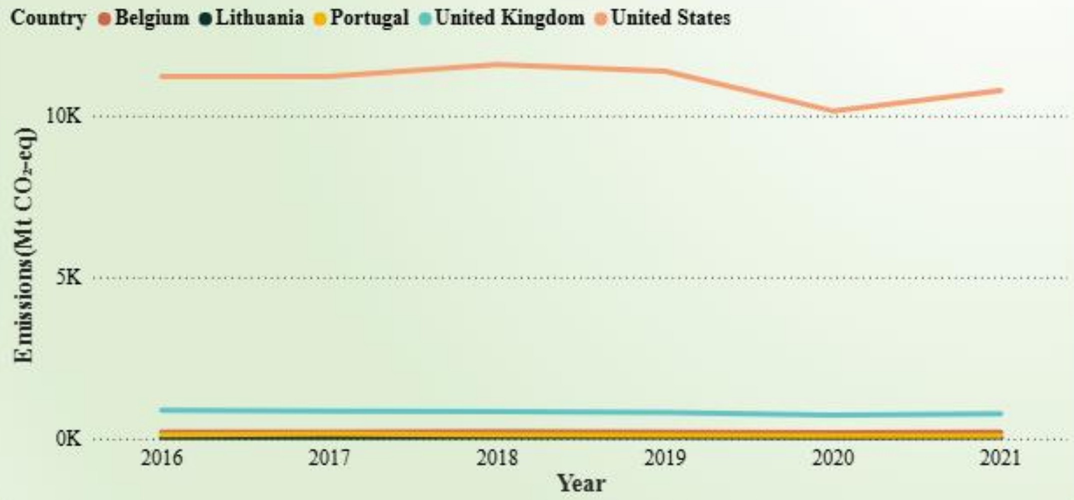
Overall Emissions(Mt CO_2-eq)

73.15K

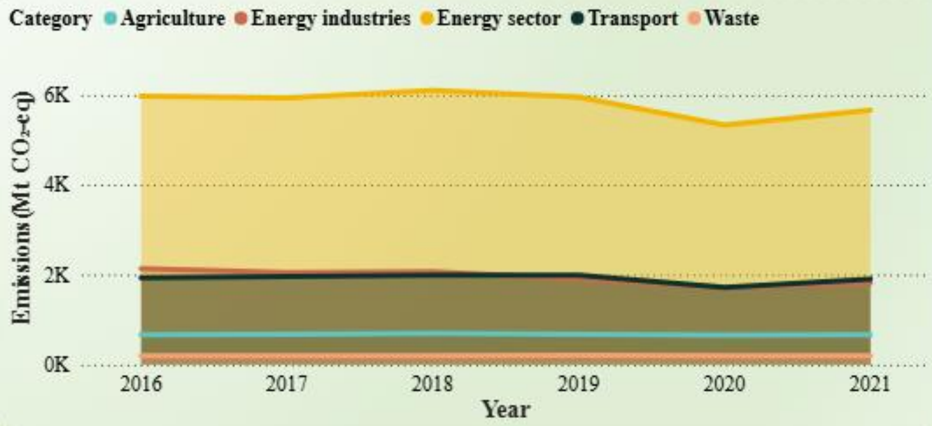
Total Emissions by Year



Emission Over Time by Country



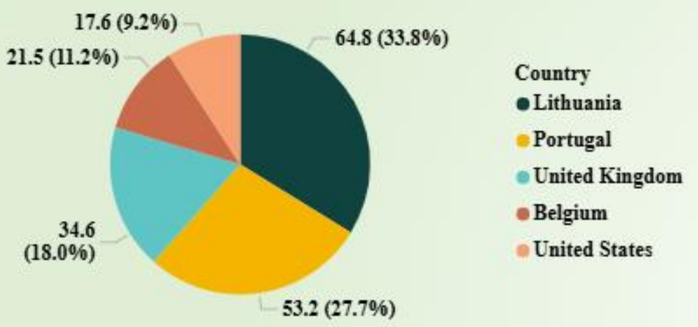
Category-wise Emissions



Powering the Transition: Renewable Energy Across Nations

Analyzing the rise of renewable energy from 2016 to 2021 and its role in driving sustainable change.

Overall Renewable Share(%)



Overall Renewable Share(%)

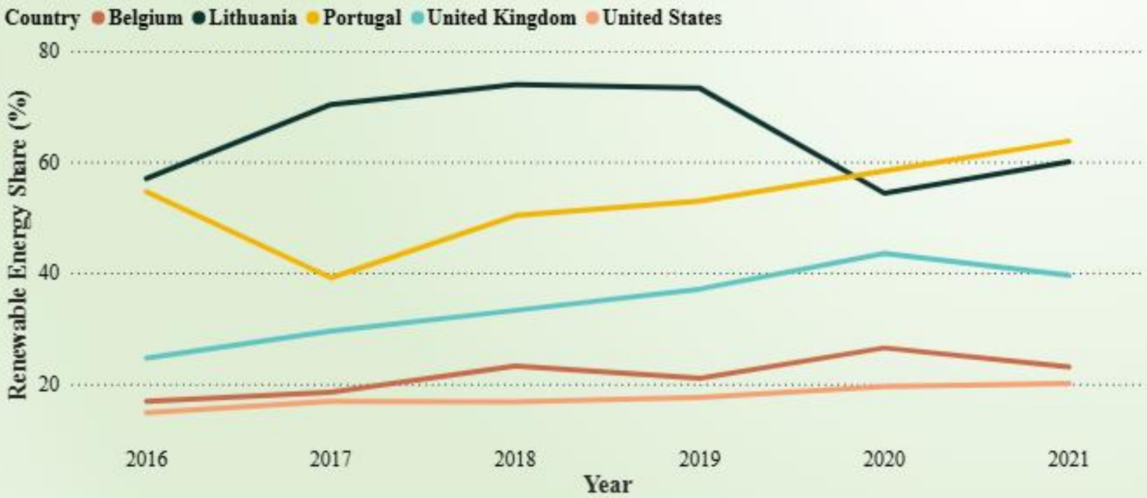
38.35

Please select a Year or Country

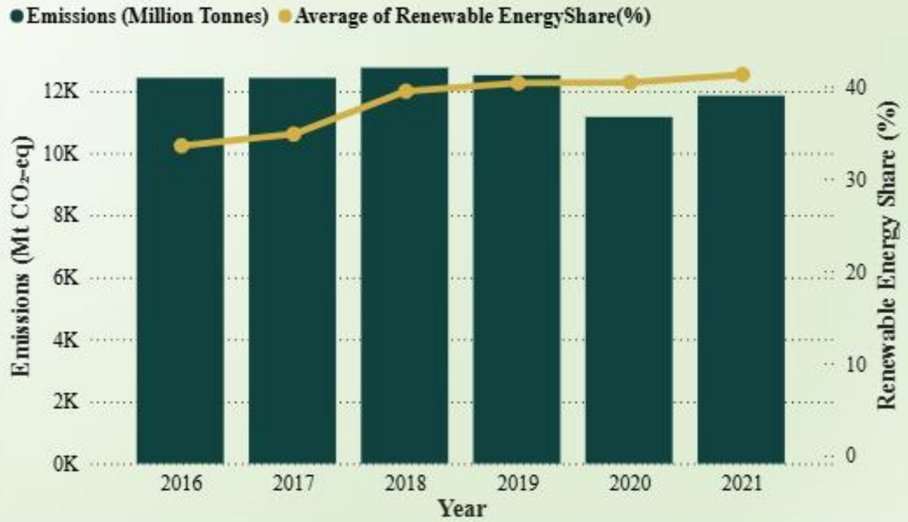
Average Renewable Share by Year

Year	Renewable Share (%)
2016	33.60
2017	34.86
2018	39.50
2019	40.38
2020	40.46
2021	41.30

Renewable Share Over Time by Country



Emissions vs Renewable Energy Adoption Over Time



Environmental Policy in Action: Who’s Leading the Transition?

Analyzing the Impact of Climate Policy on Energy Transition and Emission Outcomes

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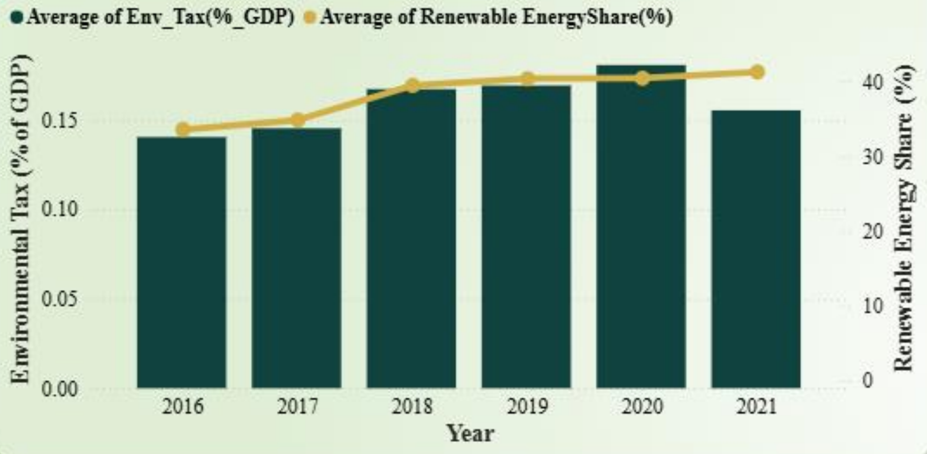
Environmental Tax and Renewable Energy Trends



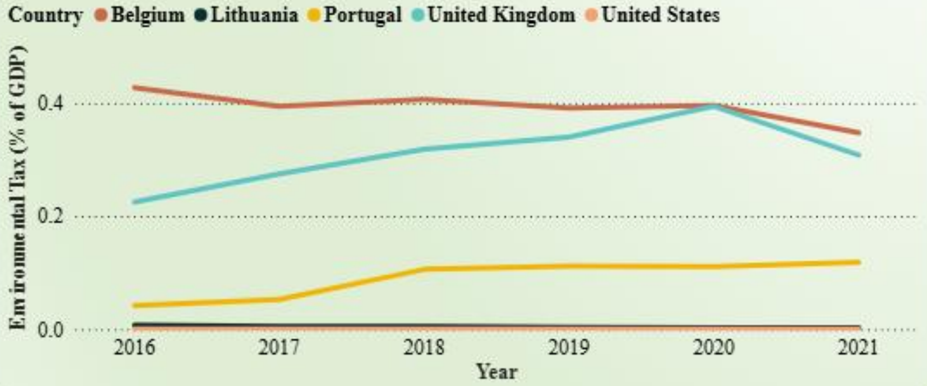
Overall Environmental Tax(% of GDP):

0.16

Sustainability Score and Ranking by Country

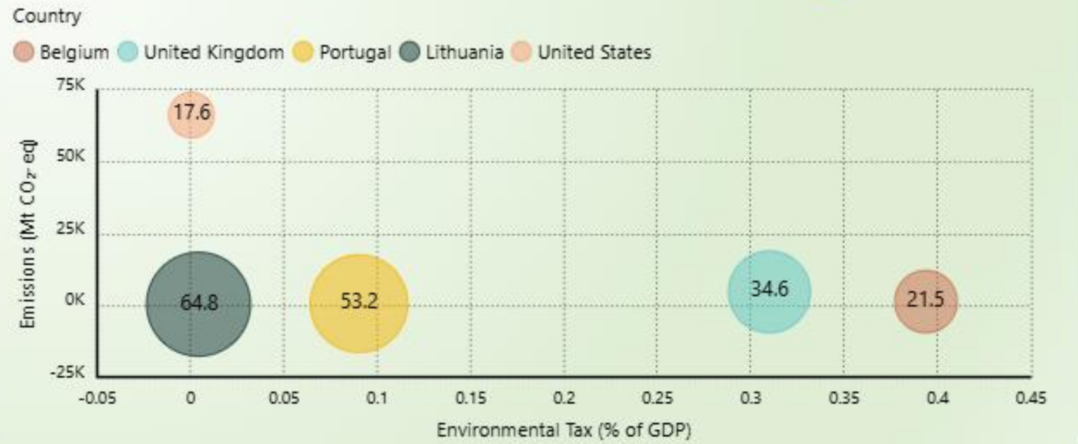
Country	Sustainability Score	Sustainability Rank
Belgium	70.10	1
Portugal	63.30	2
United Kingdom	63.10	3
Lithuania	60.20	4
United States	10.90	5

Environmental Tax Over Time by Country



Policy Effort vs Emission and Renewable Energy by Country

Renewable share (%) shown by bubble size and label; positioned by tax effort and emissions.



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Over

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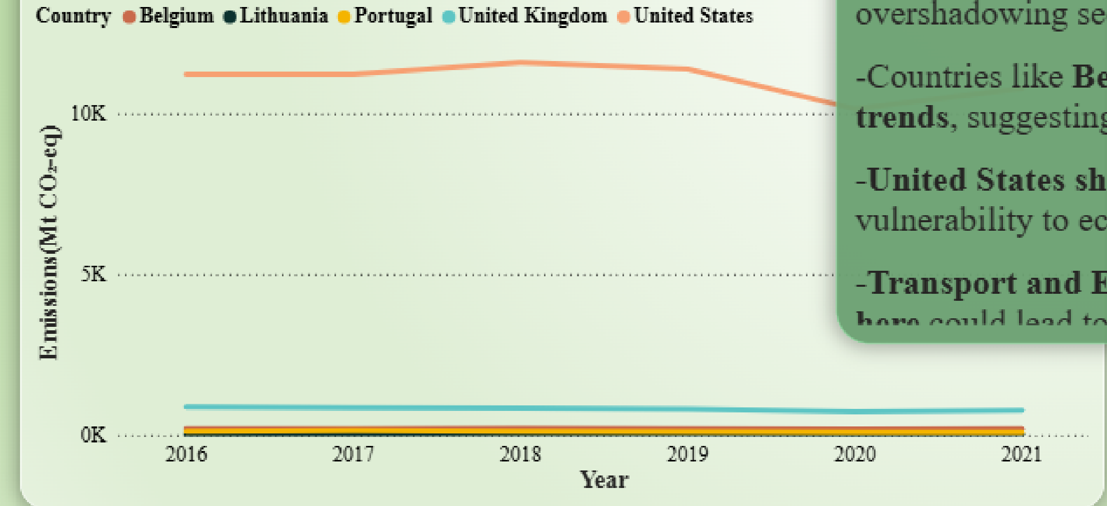
Emission Over Time by Country



Fun Facts:

-**Paris Agreement, signed in 2015**, urged countries to cut emissions, yet from 2016 to 2019, **emissions remained high and largely unchanged**, especially in the U.S.

-**The United States alone produced over 66,000 Mt CO_2-eq**, contributing **more than 90%** of total emissions among the five countries shown.

-**Emissions dipped in 2020**, especially in the U.S., likely due to the **COVID-19 pandemic's impact on transport and industry,** not long-term structural reform.

-**A slight rebound in 2021** highlights how emissions reductions were temporary and **not driven by climate policy**.

-Across all countries, the **Energy sector consistently generates the most emissions**, overshadowing sectors like Waste and Agriculture.

-Countries like **Belgium and Portugal** show **flat or gradually declining emission trends**, suggesting more stable, lower-emission profiles.

-**United States shows significant year-to-year changes**, reflecting both its size and vulnerability to economic shifts.

-**Transport and Energy** are the most impactful sectors—**even small reductions here** could lead to meaningful global climate improvements.

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Filter

Clear X

Country: All | Year: All | Category: All

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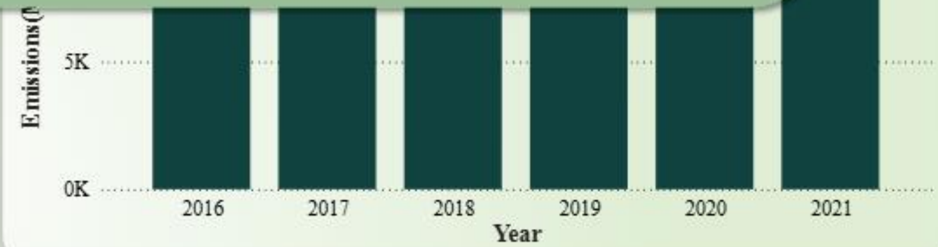
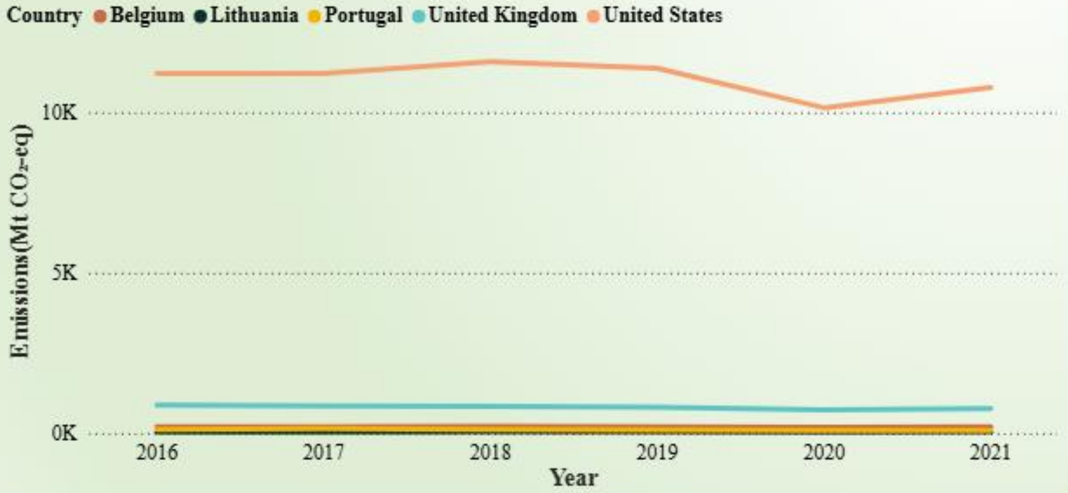


Emission Over Time by Country



Category-wise Emissions